

Isabelle A. Rose - Cover Letter for Quantum Computing Research Scientist at Google

Dear Hiring Office,

I am a theoretical physicist with a bachelor's degree in physics from MIT, soon to acquire a master's degree in theoretical physics from Oxford. I have honed my academic focus toward low-number high-resolution fundamental particle interactions, including in quantum field theory, quantum algorithms and quantum error correction. I am also trained in physics-applied machine learning. I am excited to bring my mathematical fluency to designing faulty-tolerant algorithms and methods of reducing the number of error-correcting qubits required to execute such algorithms. In my liaising with the trapped ion research group at Oxford, it seems that quantum information, machine learning hybrids are the most viable next step in quantum computing, and that there is a want for theoretical modelling and understanding of their results. It seems my skills at Google would be best suited toward this endeavour.

Best regards,
Isabelle Rose